

Duration: 3hrs**[Max Marks:80]**

- N.B. : (1) Question No 1 is Compulsory.
 (2) Attempt any three questions out of the remaining five.
 (3) All questions carry equal marks.
 (4) Assume suitable data, if required and state it clearly.

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| 1 | Attempt any FOUR | [20] |
| a | List and Draw security functional group components in IoT reference architecture | [5] |
| b | Examine how the following sensors can be used as a part of sensing technology in IoT application design 1) Ultrasonic sensors 2) Accelerometer 3) GPS sensor | [5] |
| c | Describe why connecting smart objects are different from connecting regular devices such as a laptop? | [5] |
| d | Explain features of SQL. | [5] |
| e | Compare IoT deployment level 1 and level 6. | [5] |
| 2 | a Considering smart city application explain characteristics of IoT. | [10] |
| | b Compare and contrast various phases of data analytics. | [10] |
| 3 | a Describe how LoRaWAN improves battery lifetime. | [10] |
| | b Explain various interfaces available in Raspberry Pi. | [10] |
| 4 | a Explain IoT security requirements. | [10] |
| | b Classify IoT communication models and briefly explain each. | [10] |
| 5 | a List steps involved in IoT system design methodology. Explain domain model specification for home automation application. | [10] |
| | b Explain MQTT protocol with an example highlighting functions of broker. | [10] |
| 6 | a Describe various cloud service models. | [10] |
| | b Draw the layered attacker model and explain possible attacks in IoT/M2M | [10] |